

Identifying an Optimal Adjuvant Radiotherapy Dose for Extremity and Trunk Merkel Cell Carcinoma Following Resection

An Analysis of the National Cancer Database

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IMPORTANCE Merkel cell carcinoma (MCC) is a rare and aggressive cutaneous neuroendocrine neoplasm with a high risk of recurrence following resection. Despite a rising incidence over the past 3 decades, there is a paucity of prospective data owing to the rarity of this disease.

OBJECTIVE To determine the optimal adjuvant radiation therapy (RT) dose following resection of localized MCC of the extremities or trunk.

DESIGN, SETTING, AND PARTICIPANTS Using the National Cancer Database, a large national database consisting of a heterogeneous population and treatment settings, we retrospectively analyzed a cohort of 2093 patients 18 years or older with stage I to III MCC of the extremities and/or trunk treated with definitive surgery and adjuvant RT between 1998 and 2011. Exclusion criteria included receiving treatment with palliative intent, preoperative RT, non-external-beam RT, and radiation dose of 30 Gy or lower or 70 Gy or higher. Cox proportional hazards regression model was used to compare overall survival (OS) between RT dose groups, accounting for age, sex, race, stage, surgery type, margin status, comorbidities, and use of chemotherapy.

EXPOSURES Radiation therapy dose was categorized into 4 groups: group 1 received the lowest dose (>30 to <40 Gy); group 2, the next lowest (40 to <50 Gy); group 3, the second highest dose (50 to 55 Gy); and group 4, the highest dose (>55 to 70 Gy).

MAIN OUTCOME AND MEASURE Overall survival.

RESULTS Data from 2093 patients were analyzed; there were 1293 men (61.8%) and 800 women (38.2%) (median age, 73 years). After a median follow-up of 37 months for the entire cohort, 904 deaths were reported. The 3-year OS rates for groups 1, 2, 3, and 4 were 41.8%, 69%, 69.2%, and 66%, respectively (omnibus $P < .001$). Compared with group 3 (50 to 55 Gy), equivalent OS was seen in group 2 (40 to <50 Gy; adjusted hazard ratio [AHR], 0.89; 95% CI, 0.63-1.27; $P = .52$) and group 4 (>55 to 70 Gy; AHR, 1.18; 95% CI, 0.93-1.48; $P = .17$), but worse OS was found in group 1 (>30 to <40 Gy; AHR, 2.63; 95% CI, 1.44-4.80; $P < .001$).

CONCLUSIONS AND RELEVANCE Adjuvant RT dose from 40 to lower than 50 Gy appears adequate for extremities and/or trunk stage I to III MCC, with OS equivalent to that found at higher-dose regimens (>50 to 70 Gy).

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Merkel cell carcinoma (MCC) is a rare and aggressive neuroendocrine neoplasm of the skin, with approximately 1600 new cases reported in the United States per year.¹⁻³ The incidence increased 3-fold from the 1980s to 2000s, which is speculated to be owing to an increasing elderly population with substantial prior sun exposure, improved diagnostic technique (ie, cytokeratine-20 immunohistochemical analysis), and higher prevalence of T-cell immunosuppression (eg, acquired immunodeficiency syndrome, solid organ transplantation).³⁻⁸

For localized MCC, wide local excision with sentinel lymph node biopsy (SLNB) is the mainstay of treatment for operative candidates.⁹⁻¹⁵ Adjuvant radiation therapy (RT) to the primary tumor site with or without regional lymph nodes (LNs) is considered for larger tumors (≥ 2 cm), positive or close resection margins, the presence of lymphovascular invasion, the lack of SLNB, or an immunocompromised patient.¹⁶⁻¹⁹ For LN-positive disease, complete LN dissection, nodal RT, or both is considered.^{15,20} Owing to high rates of treatment resistance and recurrence, survival rates for MCC are inferior to those for melanoma; the reported 5-year survival rates for patients with localized and nodal MCC at presentation are approximately 65% and 40%, respectively.^{4,11} The addition of chemotherapy has not consistently shown a survival benefit for stage I to III MCC.^{4,21,22}

The optimal RT dose has not been clearly defined, and the National Comprehensive Cancer Network suggests a broad range of doses ranging from 50 to 66 Gy, based on amount of residual disease after resection.¹⁵ Practice patterns vary, given the rarity of MCC, with evidence primarily from single-institution retrospective studies.^{19,23-26} These studies consistently show a dose-response for locoregional disease control (LRC); however, whether an improvement in LRC translates into a survival benefit has not been shown owing to small sample sizes. Therefore, the aim of this analysis was to compare survival among various adjuvant RT dose ranges for MCC using a population-based database. Given the closely confined neurovascular anatomy of the head and neck, as well as frequent cosmetic and functional surgical limitations in these sites,²⁷ this analysis was limited to patients with MCC confined to the extremities and trunk given the greater ability to achieve optimal resection at these sites.

Methods

Data Source and Patient Characteristics

The National Cancer Database (NCDB) is a clinical oncology database sourced from hospital registry data collected from more than 1500 Commission on Cancer-accredited facilities.²⁸ Only deidentified data were used, and so the Boston University School of Medicine institutional review board waived approval. Using histology code 8247 from the *International Classification of Diseases for Oncology, Third Edition*, we identified all patients 18 years or older with MCC who underwent surgery followed by adjuvant external-beam RT between 1998 and 2011. Patients with carcinoma in situ, metastatic disease,

Key Points

Question What is the appropriate adjuvant radiation dose delivered for extremity and/or trunk localized Merkel cell carcinoma following resection?

Findings In this cohort study of 2093 patients identified in the National Cancer Database, adjuvant radiotherapy dose from 40 to lower than 50 Gy achieved comparable overall survival compared with the more commonly used doses of 50 to 55 Gy and higher than 55 to 70 Gy.

Meaning For localized Merkel cell carcinoma of the extremities and/or trunk treated with curative surgery and adjuvant radiotherapy, doses from 40 to less than 50 Gy may be appropriate to achieve survival equivalent to that achieved with higher doses and perhaps with less toxic effect.

missing surgical or RT treatment information, and non-external-beam radiotherapy were excluded.

To account for potential confounders to the natural history or treatment selection of MCC, patient and demographic variables obtained included year of diagnosis, age, sex, race (white or nonwhite, Hispanic or non-Hispanic), insurance status (private, Medicaid or other government, Medicare, or uninsured), and treatment facility type (community program, comprehensive community program, or academic program). Clinical and pathologic characteristics included Charlson-Deyo comorbidity score²⁹ (available only from 2002 onwards), AJCC stage (American Joint Committee on Cancer),³⁰ surgery type (local, wide, or radical), tumor size, LN status, margin status, and use of adjuvant chemotherapy. Tumor size was categorized into 2 cm or smaller, larger than 2 to 3 cm, larger than 3 to 4 cm, and larger than 4 cm. The LN status was categorized into 0, 1, or 2 or more positive LNs. Margin status was categorized into negative (no residual tumor) or positive (residual tumor at either gross or microscopic examination of resected specimen, not otherwise specified). Total radiation dose delivered was calculated as the sum of regional and boost dose and was categorized into 4 dose groups: group 1 was the lowest-dose group (>30 to <40 Gy); group 2, second lowest (40 to <50 Gy); group 3, second highest (50 to 55 Gy); and group 4, the highest dose (>55 to 70 Gy). Any patient treated with a dose of 30 Gy or lower or higher than 70 Gy ($n = 166$) was excluded because these dose ranges are often used for palliative and nonstandard MCC treatments, respectively. Group 3, the dose range 50 to 55 Gy, was chosen as the reference group because these doses have consistently been used in prior single-institution studies and been shown to provide significant benefit in LRC following surgery.²⁴⁻²⁶

Statistical Analysis

The primary outcome of interest was overall survival (OS) from the time of diagnosis. The nonparametric Wilcoxon-Mann-Whitney test was used to assess difference in median age among the dose groups. The χ^2 test (Fisher exact test for small samples) was used to compare categorical variables (patient, clinical, and pathologic characteristics) among the dose groups. Median OS and 3-year OS rates were estimated

Table 1. Patient and Demographic Characteristics for Overall Cohort and Each Radiation Dose Group^a

Characteristic	All Patients (n = 2093)	Dose Group				P Value ^b
		>30-<40 Gy (n = 61)	40-<50 Gy (n = 263)	50-55 Gy (n = 1104)	>55-70 Gy (n = 665)	
Age, median (IQR), y	73 (16)	76 (13)	74 (18)	73 (15)	72 (17)	.01
Sex						.02
Men	1293 (61.8)	37 (60.7)	140 (53.2)	695 (63.0)	421 (63.3)	
Women	800 (38.2)	24 (39.3)	123 (46.8)	409 (37.1)	244 (36.7)	
Race						.61
White	2027 (96.9)	58 (95.1)	254 (96.6)	1074 (97.3)	641 (96.4)	
Nonwhite	66 (3.2)	3 (4.9)	9 (3.4)	30 (2.7)	24 (3.6)	
Hispanic origin						.43
Non-Hispanic	1902 (90.9)	54 (88.5)	243 (92.4)	1009 (91.4)	596 (89.6)	
Hispanic	191 (9.1)	7 (11.5)	20 (7.6)	95 (8.6)	69 (10.4)	
Insurance status						.46
Private	633 (30.2)	11 (18.0)	87 (33.1)	324 (29.4)	211 (31.7)	
Medicaid or other govt	49 (2.3)	2 (3.3)	6 (2.3)	29 (2.6)	12 (1.8)	
Medicare	1370 (65.5)	47 (77.1)	166 (63.1)	731 (66.2)	426 (64.1)	
Uninsured or unknown	41 (2.0)	1 (1.6)	4 (1.5)	20 (1.8)	16 (2.4)	
Facility type						.29
Community	204 (9.8)	6 (9.8)	29 (11.0)	102 (9.2)	67 (10.1)	
Comp community	1187 (56.7)	36 (59.0)	134 (51.0)	621 (56.3)	396 (59.6)	
Academic	702 (33.5)	19 (31.2)	100 (38.0)	381 (34.5)	202 (30.4)	

Abbreviations: Comp, comprehensive; govt, government-provided insurance; IQR, interquartile range.

^a Unless otherwise indicated, data are reported as number (percentage) of patients.

^b P value obtained by χ^2 or Fisher exact test for categorical variables to measure the distribution of each characteristic among the 4 dose groups.

using the Kaplan-Meier method, and log-rank tests were used to determine statistical significance. Cox proportional hazards regression modeling was used to compute crude and adjusted hazard ratios (HR) for death with 95% confidence intervals (CIs). All potential confounders including age, sex, race, insurance status, treatment facility, Charlson-Deyo comorbidity score, AJCC stage, surgery type, margin status, tumor size, LN status, and chemotherapy use were included in the multivariate analysis. All tests were 2-sided, and $P < .05$ was considered statistically significant. All statistical computations were performed on SAS software, version 9.4 (SAS Institute Inc) or GraphPad Prism software, version 3.0 (GraphPad Software).

Results

Patient Characteristics

The final eligible study cohort consisted of 2093 patients with nonmetastatic MCC treated definitively with surgery and adjuvant RT: 40.3% were stage I (n = 703); 24.4% were stage II (n = 425); and 35.4% were stage III (n = 617). All patients underwent initial surgery, with 59.8% receiving radical resection (n = 1226), 15.9% receiving wide excision (n = 325), and 24.4% receiving local excision (n = 500). Negative margins were obtained in 87.5% of cases (n = 1763), with the remaining 12.5% containing microscopic or gross residual tumor (n = 252). Only 18.2% received adjuvant chemotherapy in addition to RT fol-

lowing surgery (n = 380). All patient and demographic characteristics are summarized in Table 1. All clinical and pathologic characteristics are summarized in Table 2.

Radiotherapy Dose Utilization Trends

The Figure, A, illustrates use of the 4 RT groups from 1998 to 2011. In 2011, most patients were treated with a group 3 dose (50 to 55 Gy; 52.23%; n = 129), followed by group 4 (>55 to 70 Gy; 36.44%; n = 90), group 2 (40 to <50 Gy; 8.50%; n = 21), and finally group 1 (>30 to <40 Gy; 2.83%; n = 7). Notably, the use of group 4 RT doses (>55 to 70 Gy) had an absolute 13.77% increase in use from 2003 to 2011. Conversely, the use of group 2 doses (40 to <50 Gy) declined by an absolute 13.5% from 2003 to 2011. The use of group 1 doses (>30 to <40 Gy; 0.5% absolute decrease) and group 3 doses (50 to 55 Gy; 0.23% absolute increase) remained relatively stable during this period.

As summarized in Table 1, median age and number of men compared with number of women were not similar among the 4 radiation dose subgroups. Specifically, patients were slightly older in the lowest-dose group 1 (>30 to <40 Gy), with a median (interquartile range) age of 76 (13) years compared with 74 (18), 73 (15), and 72 (17) years for the higher-dose groups 2, 3 and 4, respectively ($P = .01$). As summarized in Table 2, the number of patients receiving a radical resection (vs wide or local excision), obtaining negative surgical margins, and receiving adjuvant chemotherapy differed among the 4 radiation dose groups as well. Specifically, more patients received a radical resection with subsequent negative margins in dose groups 2

Table 2. Clinical and Pathologic Characteristics of the Overall Cohort and Each Radiation Dose Group^a

Characteristic ^b	All Patients (n = 2093)	Dose Group				P Value ^c
		>30-<40 Gy (n = 61)	40-<50 Gy (n = 263)	50-55 Gy (n = 1104)	>55-70 Gy (n = 665)	
Charlson-Deyo score						.70
0	1516 (79.8)	39 (78.0)	176 (79.3)	808 (81.0)	493 (78.4)	
1	301 (15.9)	8 (16.0)	35 (15.8)	155 (15.5)	103 (16.4)	
2	82 (4.3)	3 (6.0)	11 (5.0)	35 (3.5)	33 (5.3)	
Tumor size, cm						.06
≤2	957 (55.9)	17 (40.5)	113 (54.6)	535 (58.2)	292 (53.8)	
>2 to ≤3	357 (20.9)	10 (23.8)	39 (18.8)	198 (21.6)	110 (20.3)	
>3 to ≤4	147 (8.6)	7 (16.7)	21 (10.1)	74 (8.1)	45 (8.3)	
>4	250 (14.6)	8 (19.1)	34 (16.4)	112 (12.2)	96 (17.7)	
Positive lymph nodes						.66
0	833 (55.2)	13 (40.6)	101 (56.7)	462 (54.7)	257 (56.5)	
1	341 (22.6)	9 (28.1)	40 (22.5)	197 (23.3)	95 (20.9)	
≥2	335 (22.2)	10 (31.3)	37 (20.8)	185 (21.9)	103 (22.6)	
AJCC stage						.29
I	703 (40.3)	15 (34.9)	94 (44.1)	386 (40.9)	208 (38.1)	
II	425 (24.4)	12 (27.9)	49 (23.0)	212 (22.5)	152 (27.8)	
III	617 (35.4)	16 (37.2)	70 (32.9)	345 (36.6)	186 (34.1)	
Surgical procedure						.004
Local	500 (24.4)	22 (36.7)	63 (24.5)	234 (21.5)	181 (28.1)	
Wide	325 (15.9)	10 (16.7)	33 (12.8)	173 (15.9)	109 (16.9)	
Radical	1226 (59.8)	28 (46.7)	161 (62.7)	683 (62.7)	354 (55.0)	
Surgical margins						<.001
Negative	1763 (87.5)	42 (73.7)	216 (85.0)	979 (91.8)	526 (82.6)	
Residual tumor, NOS	252 (12.5)	15 (26.3)	38 (15.0)	88 (8.3)	111 (17.4)	
Adjuvant chemotherapy						.05
No	1713 (81.8)	44 (72.1)	210 (79.9)	924 (83.7)	535 (80.5)	
Yes	380 (18.2)	17 (27.9)	53 (20.2)	180 (16.3)	130 (19.6)	

Abbreviations: AJCC, American Joint Committee on Cancer; NOS, not otherwise specified.

^a Unless otherwise indicated, data are reported as number (percentage) of patients.

^b Missing data: The Charlson-Deyo score was missing for 194 patients (9.3%);

tumor size missing for 382 (18.3%); lymph node status missing for 584 (27.9%); AJCC stage missing for 348 (16.7%); surgical margins missing for 78 (3.7%); and surgical procedure missing for 42 (2.0%).

^c P value obtained by χ^2 or Fisher exact test to measure the distribution of each characteristic among the 4 subgroups.

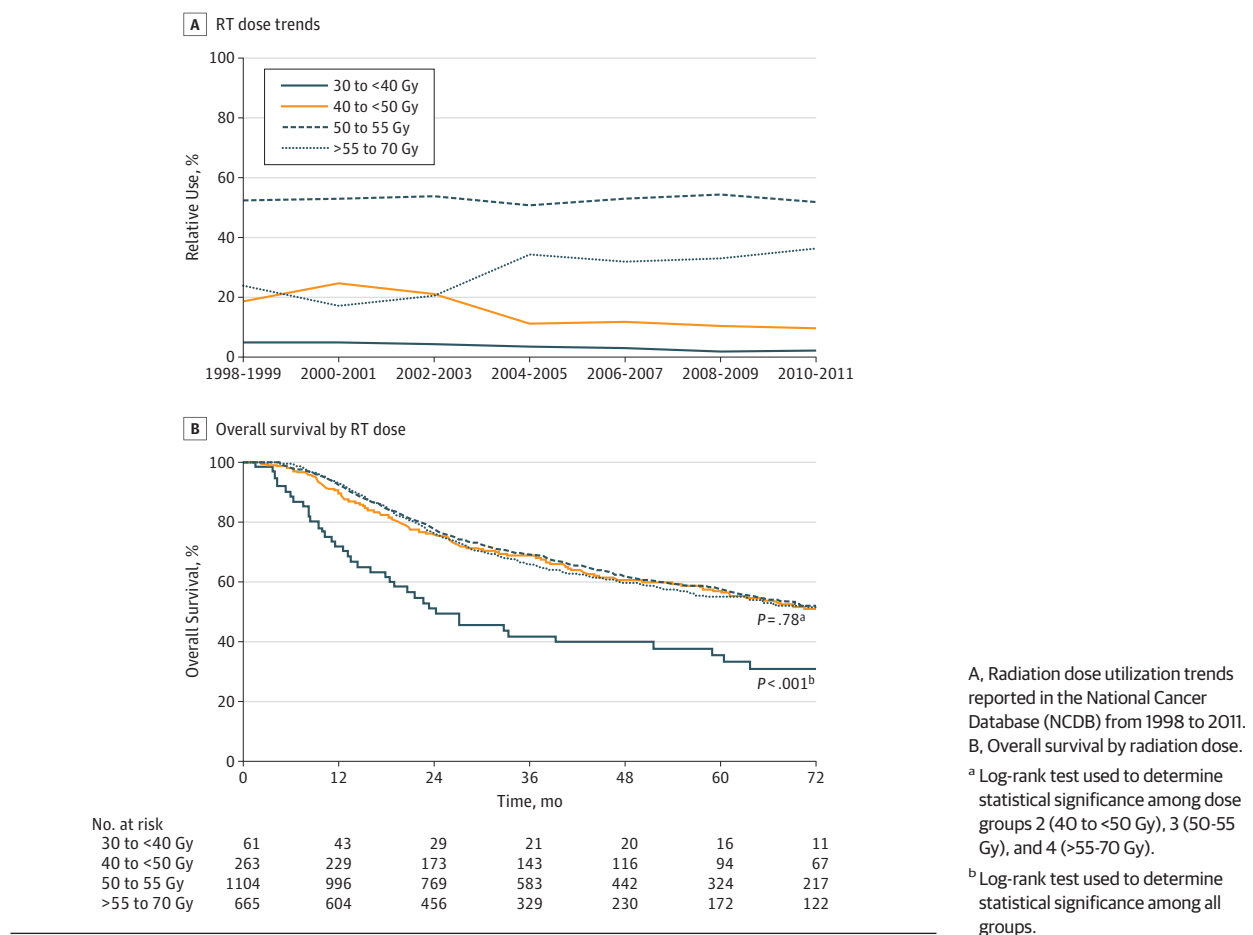
and 3 (40 to <50 Gy and 50 to 55 Gy) than in the other dose groups. More patients in dose group 1 (>30 to <40 Gy) received adjuvant chemotherapy (concurrently or sequentially) than in the other dose groups. There was also a trend in the difference in tumor size among the 4 dose groups, with patients in group 3 (50 to 55 Gy) having more tumors 2 cm or smaller while those in group 1 (>30 to <40 Gy) had more tumors larger than 4 cm.

Survival Analysis By Radiotherapy Dose

After a median follow-up of 37 months for the entire cohort (51.7 months for surviving patients), 904 patients had died following surgery and adjuvant RT. The median survival time and 3-year OS rate for the entire cohort were 74.4 months and 67.4%, respectively. Table 3 summarizes the median survival time and 1-, 2-, and 3-year OS rates for the entire cohort and within each RT dose group. Median survival time was significantly lower (24.3 months) in the lowest-dose group than in the 3 higher-dose groups (80.0, 75.4, and 75.3 months for

groups 2, 3, and 4, respectively) ($P < .001$). As seen in the Figure, B, survival rates were lower in group 1 than in the other 3 groups, although there was no difference in survival rates between 3 highest dose groups.

Compared with the patients in group 3 (50 to 55 Gy), patients in group 1 (>30 to <40 Gy) had a significantly higher risk of death as found by multivariate analysis and adjusting for all potential confounders (adjusted HR, 2.63; 95% CI, 1.44-4.80; $P < .001$). Within this multivariate model, there was not a significant difference in mortality risk with RT doses between group 3 and group 2 (40 to <50 Gy; adjusted HR, 0.89; 95% CI, 0.63-1.27; $P = .52$) or group 3 and group 4 (>55 to 70 Gy; adjusted HR, 1.18; 95% CI, 0.93-1.48; $P = .17$). When group 2 (40 to <50 Gy) was set as the referent, the lack of difference in mortality risk compared with group 3 (50 to 55 Gy; adjusted HR, 1.12; 95% CI, 0.79-1.60; $P = .52$) and group 4 (>55 to 70 Gy; adjusted HR, 1.32; 95% CI, 0.91-1.92; $P = .15$) was confirmed. These results are summarized in Table 4.

Figure. Doses Used for Adjuvant Radiation Therapy (RT) in Patients With Surgically Resected Merkel Cell Carcinoma of the Extremities and/or Trunk**Table 3. Survival Rates for Study Cohort and By Radiation Dose**

Dose Group	Patients, No.	Deaths	Median Survival, mo	Actuarial Survival, %			P Value
				1-Year	2-Year	3-Year	
All patients	2093	904	74.4	91.6	76.2	67.4	NA
1 (>30 to <40 Gy)	61	41	24.3	71.9	51.4	41.8	<.001
2 (40 to <50 Gy)	263	119	80.0	89.5	75.8	69.0	
3 (50 to 55 Gy)	1104	463	75.4	92.4	77.7	69.2	
4 (>55 to 70 Gy)	665	281	75.3	92.8	76.3	66.0	

Discussion

Even with optimal resection of MCC, locoregional and distant recurrence rate has been reported as high as 79%, which explains the poor 5-year survival even for localized disease.^{4,31} There remains debate regarding the routine inclusion of adjuvant therapy following resection.^{9,32,33} However, several single-institution studies have shown an LRC benefit with the use of adjuvant RT.^{19,24-26} Furthermore, analyses from a population-based database have shown a survival benefit with the addition of postoperative RT for stage I to III MCC.^{4,18,23}

In our analysis using the NCDB, we found that for extremity or trunk MCC, patients who received adjuvant RT at doses

ranging from 40 to less than 50 Gy showed survival rates equivalent to those patients who received RT dose ranges of 50 to 55 Gy and higher than 55 to 70 Gy. These results persisted on multivariate analysis accounting for patient age, sex, tumor stage, resection type, margin status, medical comorbidities, and chemotherapy use. There was a significantly lower survival seen at adjuvant RT doses ranging from higher than 30 to lower than 40 Gy compared with all other dose ranges, after adjusting for all potential confounders. These findings suggest that, although MCC is a radiosensitive neuroendocrine tumor, RT doses in this lower range may be inadequate to eradicate marginal, in-transit, or micrometastatic disease often present after resection, which factors taken together may account for the notoriously high propensity for locoregional

Table 4. Multivariate Analysis of Overall Survival Using 2 Different Reference Groups

Dose Group	Patients, No.	Deaths	Crude HR (95% CI)	P Value	Adjusted HR (95% CI) ^a	P Value
Dose Group 3 as Reference Group						
1 (>30 to <40 Gy)	18	12	2.73 (1.53-4.89)	.001	2.63 (1.44-4.80)	<.001
2 (40 to <50 Gy)	115	40	1.01 (0.72-1.42)	.94	0.89 (0.63-1.27)	.52
3 (50 to 55 Gy)	605	207	1 [Reference]		1 [Reference]	
4 (>55 to 70 Gy)	322	119	1.15 (0.91-1.44)	.24	1.18 (0.93-1.48)	.17
Dose Group 2 as Reference Group						
1 (>30 to <40 Gy)	18	12	2.70 (1.41-5.14)	.003	2.95 (1.50-5.80)	.002
2 (40 to <50 Gy)	115	40	1 [Reference]		1 [Reference]	
3 (50 to 55 Gy)	605	207	0.99 (0.70-1.39)	.94	1.12 (0.79-1.60)	.52
4 (>55 to 70 Gy)	322	119	1.13 (0.79-1.62)	.50	1.32 (0.91-1.92)	.15

Abbreviations: AJCC, American Joint Committee on Cancer; HR, hazard ratio.

^a Multivariable model adjusting for age, gender, race, insurance status, treatment facility, Charlson-Deyo Score, surgery type, tumor size, positive

lymph nodes, AJCC stage, surgical margins, and adjuvant chemotherapy. Only patients with complete covariate data were included.

and distant recurrence even for a localized MCC tumor. The poorer outcomes observed with lower RT doses cannot be attributed to potential palliative treatment because patients with metastatic disease or designated as receiving palliative treatment were excluded from this analysis. Furthermore, patients receiving a total dose of 30 Gy were excluded, since this is a commonly used prescription for palliative treatments.

We observed an increase in the use of group 4 radiation doses (>55 to 70 Gy) for extremity and/or trunk MCC when the RT was delivered in the postoperative setting, which has been especially pronounced since 2002. During the same period, there has been a decline in the use of group 2 doses (>40 to 50 Gy) and relatively stable use of doses 1 and 3 (>30 to <40 Gy and 50 to 55 Gy). These findings are perhaps correlative with the publication of data revealing the rising incidence of MCC and the poor survival rates even for patients with localized disease. However, the results from the present analysis suggest that we should reevaluate the role of routine dose escalation in patients with localized MCC and support the use of a more modest RT dose in the appropriate scenarios. Given potential adverse effects of higher doses, particularly lymphedema risk with regional nodal irradiation, and health care expenditures associated with protracted RT,³⁴ every attempt at deescalating therapy without compromise in cancer control should be evaluated. Furthermore, the incidence of MCC is predominantly seen in an aging population, which was also observed in the present cohort, which had a median age of 73 years^{2,3}; therefore, intensifying treatment in the adjuvant setting for older patients, especially those with comorbidities or with few if any high-risk tumor features, does not seem warranted.

A similar analysis conducted by our group in patients with MCC confined to the head and neck found that RT doses of 50 to 55 Gy had significantly improved OS compared with both lower and higher dose ranges.³⁵ In contrast to patients with MCC of the head and neck region, for whom doses below 50 Gy were found to be suboptimal, patients with extremity and/or trunk MCC had equivalent OS with adjuvant RT doses of 40 to 50 Gy, perhaps owing to the surgeon's ability to obtain wider negative margins or to the presence of fewer high-risk tumor features (given the less complex neurovascular and

lymphatic anatomy in the trunk and extremity region compared with the head and neck areas).²⁷ Only crude surgical margin data and limited histological characteristics are routinely available in the NCDB, so a potential interaction between detailed pathologic features and RT dose cannot be fully assessed. Regardless, the equivalence in survival among groups 2, 3, and 4 (40 to <50 Gy, 50 to 55 Gy, and >55 to 70 Gy, respectively) remained after multivariate analysis and adjustment for all possible clinical and pathologic confounders. Likewise, the decrement in survival with group 1 doses (>30 to <40 Gy) cannot be attributed to the prevalence of high-risk tumor features, as can be inferred from the data reported in Table 2, since these results remained on multivariate analysis accounting for each of these covariates.

The results of our study should be reviewed in the context of prior studies evaluating the role of RT dose for MCC. Morrison et al¹⁹ established an effective dose-response at 46 to 50 Gy in the postoperative setting for subclinical disease in a 31-patient cohort of nonmetastatic primary or locoregionally recurrent MCC (30% extremity and/or trunk) treated at the MD Anderson Cancer Center. There were only 3 marginal recurrences and 1 in-field recurrence in the head and neck after reirradiation. Hui et al²⁵ reviewed 176 patients with nonmetastatic MCC (44% extremity and/or trunk) treated with wide local excision and adjuvant RT at the Peter MacCallum Cancer Centre and demonstrated that doses of 45 Gy or higher were predictive of LRC on multivariate analysis. Five-year locoregional recurrence-free survival was 80% for an RT dose of 45 Gy or higher vs 71% for a dose lower than 45 Gy. Foote et al²⁴ reviewed 112 MCC cases (37% on the extremity and/or trunk) treated with adjuvant RT at the University of Queensland and revealed optimal LRC with RT doses of 50 Gy or higher for subclinical disease and 55 Gy or higher for gross disease. The in-field relapse rate was only 3%, but there were no in-field relapses with doses of 50 Gy or higher and 55 Gy or higher for subclinical and gross disease, respectively.

Finally, in a pooled analysis of 88 patients with MCC treated with resection and adjuvant RT doses of 50 to 60 Gy with concurrent carboplatin from 3 prospective trials conducted by the Trans Tasman Radiation Oncology Group (TROG),³⁶ no sig-

nificant differences in LRC were found between patients with negative margins vs microscopic or gross residual disease, suggesting that even doses below 50 Gy may be sufficient in the adjuvant setting for negative margins or subclinical disease. On these trial protocols, RT doses of 45 to 50 Gy were administered to clinically uninvolved, at-risk nodal regions that were outside the surgical volume. Furthermore, TROG recommended dose reductions to 45 Gy particularly for lower-extremity tumors below the knee, inguinal regions, or in locations where 50 Gy exceeded normal tissue tolerances.

Limitations

Notably, the higher dose recommendation (ie, ≥ 45 to 50 Gy) in the 3 retrospective series^{19,24,25} than that suggested by our analysis (ie, 40 to < 50 Gy) may be driven by the inclusion of head and neck primary tumors and patients with gross residual disease following resection. Furthermore, the primary outcome assessed in the earlier studies was LRC. The present findings suggest that RT doses of 40 to 50 Gy result in OS equivalent to that found in higher dose regimens. However, whether doses between 40 and less than 45 Gy result in equivalent LRC compared with higher dose regimens is unknown; the NCDB is limited by lack of LRC data.

Other limitations of our study are worth noting. First, Charlson-Deyo comorbidity index was only available within the NCDB beginning in 2002; therefore, all patients treated between 1998 and 2001 did not have comorbidity scores. Second, our analysis assumes that the total dose was delivered at the conventional 2 Gy per fraction based on the elapsed treatment days. We did not examine dose per fraction to account for all the possible RT dose fractionation schemes used, which

may result in varying radiobiological effective doses. Third, the NCDB does not provide detailed data on radiotherapy technique (eg, megavoltage photons, electrons, orthovoltage) and field designs, which would indicate the amount of the primary tumor and/or regional LNs that were treated. Likewise, details regarding the extent of wide or local excision (eg, Mohs vs other techniques) were not included in this analysis. However, tumor size, crude margin status, and AJCC stage were all accounted for in the multivariate model; therefore, these specific data unavailable in the NCDB would be unlikely to significantly alter our findings.

Conclusions

In conclusion, we believe that our study constitutes the largest series of patients with extremity and/or trunk MCC from a population-based database demonstrating a relationship between adjuvant RT dose and survival. Our multivariate analysis showed a survival detriment with doses below 40 Gy; however, there was no difference in OS for doses between 40 and lower than 50 Gy vs the more commonly prescribed doses of 50 to 55 Gy or higher than 55 to 70 Gy. Since MCC is a radio-sensitive small cell cancer, routine adjuvant dose escalation above 50 Gy does not appear necessary. In the context of prior studies, our findings support doses from 40 to lower than 50 Gy from an OS end point, but it would require further study to determine whether doses of 40 to lower than 45 Gy are sufficient for LRC. These results should inform treatment recommendations for patients eligible for adjuvant RT following resection of an extremity or trunk MCC primary tumor.

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